CADT IDT វិទ្យាស្ថានបច្ចេកវិទ្យាឌីជីថល Institute of Digital Technology

Faculty of Digital Engineering
Department of Computer Science
Specialized in Software Engineering

Capstone Proposal

Student Name	*Chen Sovanpanha, Phally Sovanratana, Chao Vanchhoeurng, Yann Sokmeng, Thy Sereythi, Kim Seyha*
Group	*28*
Project Title	**Smart Plant Companion & AI Health Scanner**
Started Date	*14/10/2026*
Ended Date	*31/03/2027*

1. Project Description

1.1 Background and target users

This project is for people who enjoy growing plants at home, on balconies, or in small gardens. It focuses mainly on indoor ornamental plants and hobby gardening, with a few common edible plants included if they fit the project.

1.2 Problem statement

Beginners may be unsure how much light or water a plant needs, where to find reliable care advice, or what to do when a plant looks unhealthy. They may also forget care tasks or lose track of their growing plans.

Main problems: finding clear care information, remembering care tasks, and understanding possible plant health problems.

1.3 Proposed solution

We plan to develop Smart Plant Companion & AI Health Scanner in two consecutive stages. In Capstone I, the team will build the application first as a **Telegram Mini App** and then extend it to a **mobile application** (an Android build, if time allows), with two main features: a Plant Library with care information for about 20-30 common, locally relevant plants (a proposed target, not yet completed), and a Personal Garden where users manage their own plants and record milestones, progress, and care activities. Plant information will be stored with the application, each user's garden will be stored on their own device, and community discussion will use a Telegram group. In Capstone II, the team will research, train, evaluate, and integrate an AI Health Scanner that returns a text prediction for supported plant health classes from a leaf image. This keeps plant information and planning in one place. The scanner is an educational prediction tool, not a guaranteed diagnosis. The dataset, supported classes, and model are still under research and are not yet final.

1.4 Main features

The project will focus on these features, delivered in two stages, rather than trying to support every plant or disease. Capstone I delivers the first three features; Capstone II delivers the last.

Feature	What will be delivered
Plant Library (Capstone I)	Browse, search, and view care information, including watering and light requirements, for about 20-30 common, locally relevant plants. Plant data and photos are stored with the app so it can work offline where possible (to be tested). The team will research reliable sources and record a source for each entry.
Personal Garden (Capstone I)	Add a plant from a library template with suggested start steps, milestones, and care intervals, then edit them freely. Record progress and care activities and view in-app reminders. Phone notifications, weather-based care, and automatic scheduling are outside Capstone I.
Community (Capstone I)	A link to a Telegram group where users can ask questions and share ideas. No community system will be built inside the app.
AI Health Scanner (Capstone II)	Submit a supported leaf image and receive a text prediction for selected plant and health classes, with an uncertainty message for low-confidence or unsupported images. Supported plants and classes depend on dataset and test results.

2. Project Objective

By 16 December 2026, the team aims to deliver a working Telegram Mini App with the Plant Library and Personal Garden features, with an Android build of the same application as a stretch goal (Capstone I). By 31 March 2027, the team aims to complete the AI Health Scanner and integrate it into the application (Capstone II). These targets are our current plan; the AI classes and model method will be chosen after checking the available data. Dates marked * are proposed internal targets, not official academic dates (see Section 4).

Objective	How we will check it	Target date
Capstone I: Telegram Mini App and mobile application		
Plant information	Prepare about 20-30 selected plant records with basic care information and a source recorded for each entry.	31 Oct
Plant Library	Users can browse and search the library in the Mini App and open a plant detail page to read its care information, photo, and sources.	20 Nov
Personal Garden	Users can add a plant from a template, edit its steps and dates, update milestones, and record progress and care activities.	20 Nov
Care reminders	Users can view basic in-app reminders based on their own plant or care settings.	27 Nov
Mini App testing	Test the main Plant Library and Personal Garden workflows in Telegram on a mobile device, including offline behaviour, and record known limits.	4-10 Dec
Android build	Stretch goal: build the same application as an Android app using local phone storage, if time allows.	By 8 Dec (if time allows)
Capstone II: AI Health Scanner and model training		
Dataset	Select a suitable labelled leaf-image dataset; verify class labels and usage or licensing terms; define a limited set of supported plants and classes.	By 19 Feb*
Model training	Prepare and preprocess images, split data into training, validation, and test sets, and train our own image classifier.	By 5 Mar*

Objective	How we will check it	Target date
Model evaluation	Evaluate on separate test images using accuracy, precision, recall, F1-score, and a confusion matrix; investigate uncertain and unsupported images.	By 12 Mar*
Integration	Integrate the model so a user can submit a supported leaf image and receive a text prediction.	By 19 Mar*
Final testing	Test the integrated scanner, document its limitations, and submit the Capstone II document and final presentation.	24 and 31 Mar

* Proposed internal target; not an official academic date.

3. Methodology/Activities

The team will work in short stages and test features as they are built. Capstone I concentrates on requirements, design, and mobile application development. Capstone II concentrates on dataset preparation, model training and evaluation, integration, and final testing, so that Capstone I does not depend on unfinished AI work.

Stage	Activities	Expected output
Capstone I: Telegram Mini App and mobile application		
1. Research and requirements	Review similar plant-care apps, list what users should be able to do, and research plant information sources. Begin a preliminary review of candidate AI datasets for Capstone II. Check usage terms before using data.	Feature list and research notes
2. Plant data collection	Choose about 20-30 common plants. Gather basic care facts from reliable sources, write summaries in our own words, and record where the information came from.	Initial plant records
3. App design	Sketch the main mobile-sized screens and user flow. Plan the data structure (plant records and each user's garden) and where each is stored.	Screen drafts and data plan
4. Mini App development	Build the Plant Library and Personal Garden with Flutter (web build), opened through Telegram. Plant data and images are stored with the app and each user's garden on their device. No custom backend or login is planned. Add a link to a Telegram community group.	Working Telegram Mini App
5. Testing and documentation	Test the main user flows in Telegram on a mobile device, including offline behaviour, fix important errors, and prepare the Capstone I report and demonstration.	Tested Capstone I application and report
6. Android build (stretch goal)	If time allows, build the same Flutter code as an Android app, test local storage and offline use, and consider phone notifications for reminders.	Android build (if time allows)
Capstone II: AI Health Scanner and model training		
7. Dataset preparation	Compare labelled leaf-image datasets, verify class labels and licence or usage terms, and choose a small set of supported plants and classes after advisor confirmation. Organise images into class folders.	Selected dataset and class list
8. Image preprocessing	Split images into training, validation, and test sets before augmentation; resize and normalise images to suit the chosen model; apply augmentation to training images only.	Prepared dataset

Stage	Activities	Expected output
9. Model training and evaluation	Select an image-classification approach. Transfer learning with a pretrained model such as MobileNetV2 is the proposed starting point, not a final choice. Train and improve the classifier, evaluate it on the held-out test set (accuracy, precision, recall, F1-score, confusion matrix), and examine uncertain predictions and unsupported images.	Trained model and evaluation results
10. Integration	Connect the trained model to the application, for example through a Python prediction service, so users can submit a supported leaf image and receive a text prediction with an uncertainty message for low-confidence results and a disclaimer.	Integrated AI scanner
11. Testing and reporting	Test the integrated scanner with supported and unsupported images, document its limitations, and complete the Capstone II report and presentation.	Tested scanner and final report

Planned technologies: Flutter (web build for the Telegram Mini App; Android build as a stretch goal), Telegram BotFather to register the Mini App, a free static host, and local device storage for each user's garden. No custom backend, database server, or login system is planned for Capstone I. Python is planned for AI development in Capstone II. The hosting service, storage method, AI model, dataset, and ML framework have not been selected yet; these choices, and the number of supported plants and classes, remain subject to research, testing, and advisor confirmation.

4. Project Milestone

This is the team work plan for Capstone I and Capstone II. Rows marked **Official** follow the department timeline and are fixed. All other dates are work periods planned by the team; the Capstone II work periods are proposed internal targets and may be adjusted after advisor feedback.

Period	Team activities	Expected milestone
Capstone I		
By 9 Oct	Request an advisor for the project.	**Official:** Advisor Request Deadline.
17-19 Oct	Finish the project description, objectives, methodology, and team work plan.	**Official:** Proposal submitted by 19 Oct.
20-24 Oct	Agree on requirements, review similar applications, assign task owners, and begin plant data research.	Requirements and task list ready.
20-31 Oct	Research reliable care sources and suitable APIs; collect the first plant records. Review candidate AI datasets and check labels and licenses as preparation for Capstone II.	Initial plant data and AI research notes.
23 Oct-1 Nov	Sketch screens, plan the data structure, set up Flutter and the Telegram Mini App environment, and build a simple working skeleton.	Basic design and early prototype.
2-13 Nov	Continue the prototype, demonstrate the available Mini App features, and collect feedback during the midterm presentation window.	**Official:** Mid-Term Presentation (2-13 Nov). Progress demonstrated and feedback recorded.
14-20 Nov	Complete plant browse, search, and detail pages, plus adding plants to the Personal Garden from templates and the main edit, stop, and delete actions.	Core library and garden features working.

Period	Team activities	Expected milestone
21-27 Nov	Add milestone tracking, progress and care records, and basic in-app reminders to the Personal Garden, and add the Telegram community group link.	Care features working.
28 Nov-4 Dec	Combine the Plant Library and Personal Garden in one Mini App, host it, and test the full workflow in Telegram, including offline behaviour. Begin the Android build if time allows.	Integrated Mini App.
5-8 Dec	Test the main user flows on a mobile device. Fix major issues and note known limitations. Finish the Android build if time allows.	Tested version with known limits recorded.
9-10 Dec	Finish the report and supporting documents, and check that the application and demo are ready.	Documents ready for submission.
11 Dec	Submit all required documents by the department deadline.	**Official:** All Document Submission.
12-15 Dec	Practice the presentation, check the demo on the presentation device, and fix only critical remaining issues.	Presentation ready.
16 Dec	Present and demonstrate the Capstone I Telegram Mini App (Plant Library and Personal Garden), and the Android build if completed, and explain the plan for the AI Health Scanner in Capstone II.	**Official:** Final Capstone I Presentation.
Capstone II (proceeds after the Capstone I final presentation)		
17 Dec-31 Jan	Review Capstone I feedback, confirm the AI scope with the advisor (dataset, supported plants, and classes), verify dataset licenses, and prepare the training environment.	Capstone II scope confirmed (proposed target).
1-7 Feb	Organise and preprocess the selected dataset, split it into training, validation, and test sets, and run a small baseline experiment.	Prepared dataset and baseline results (proposed target).
8-19 Feb	Present the dataset, preprocessing, and early training results and collect feedback during the midterm presentation window.	**Official:** Mid-Term Presentation (8-19 Feb).
20 Feb-5 Mar	Train and improve the image classifier, starting with the proposed transfer-learning approach.	Trained model (proposed target).
6-12 Mar	Evaluate the model on the held-out test set (accuracy, precision, recall, F1-score, confusion matrix) and investigate uncertain and unsupported images.	Evaluation results (proposed target).
13-19 Mar	Integrate the model with the application so a supported leaf image returns a text prediction, an uncertainty message where needed, and a disclaimer.	Integrated AI scanner (proposed target).
20-23 Mar	Test the integrated scanner, document its limitations, and finish the report and supporting documents.	Documents ready for submission.
24 Mar	Submit the Capstone II document by the department deadline.	**Official:** Document Submission.
25-30 Mar	Practice the presentation, check the demo on the presentation device, and fix only critical remaining issues.	Presentation ready.
31 Mar	Present and demonstrate the completed application with the AI Health Scanner, and explain its results and limitations.	**Official:** Final Capstone II Presentation.

Seen and Approved by Advisor

_____